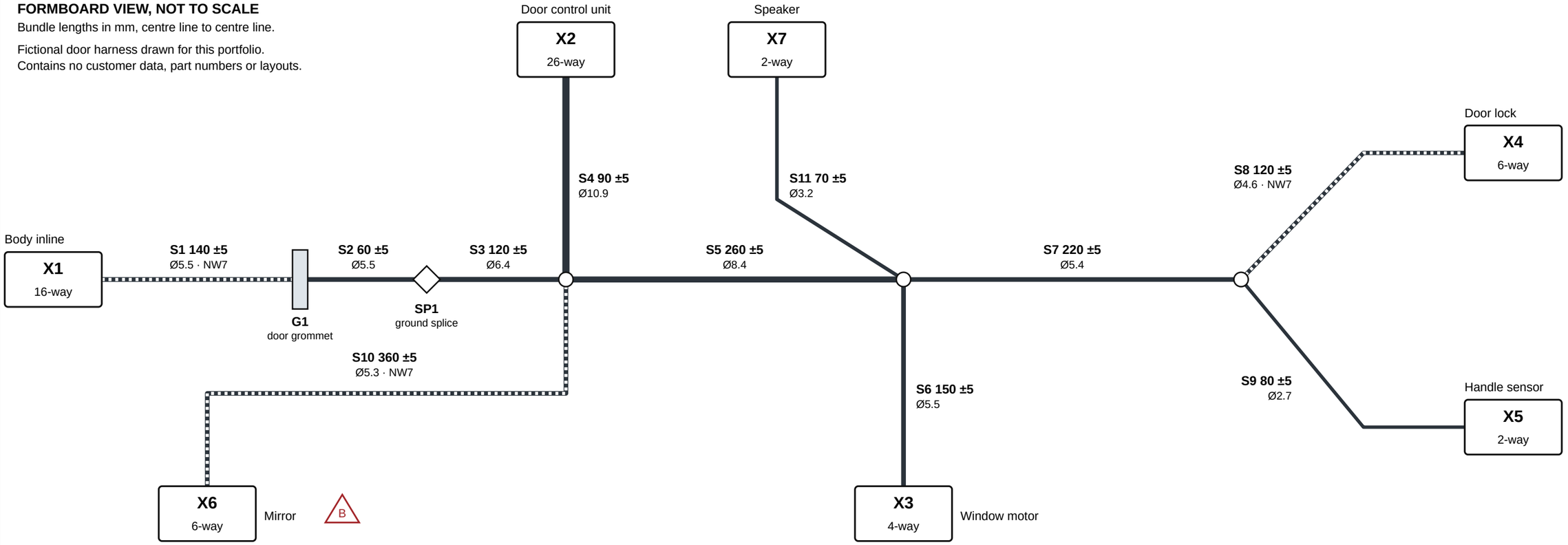
FORMBOARD VIEW, NOT TO SCALE

Bundle lengths in mm, centre line to centre line.

Fictional door harness drawn for this portfolio.
Contains no customer data, part numbers or layouts.



Wire	From	To	mm²	Colour	Function	L mm	Rev
W01	X1:1	X2:1	2.5	rt	KL30 supply	410	
W02	X1:2	SP1	2.5	br	KL31 ground	200	
W03	SP1	X2:2	2.5	br	Ground DCU	210	
W04	SP1	X5:2	0.5	br	Ground handle	680	
W05	X1:3	X2:3	0.35	ge/sw	CAN high, T1	410	
W06	X1:4	X2:4	0.35	ws/sw	CAN low, T1	410	
W07	X2:5	X3:1	2.5	gn/ws	Window motor +	500	
W08	X2:6	X3:2	2.5	gn/sw	Window motor −	500	
W09	X2:7	X3:3	0.35	bl	Hall signal	500	
W10	X2:8	X3:4	0.35	vi	Hall supply	500	
W11	X2:9	X4:1	1.5	gr/rt	Lock motor A	690	
W12	X2:10	X4:2	1.5	gr/sw	Lock motor B	690	
W13	X2:11	X4:3	0.35	or	Latch switch	690	
W14	X2:12	X4:4	0.35	or/br	Switch return	690	
W15	X2:13	X5:1	0.5	ws	Handle sensor	650	
W16	X2:14	X6:1	0.75	bl/ws	Mirror axis X	450	
W17	X2:15	X6:2	0.75	bl/sw	Mirror axis Y	450	
W18	X2:16	X6:3	0.75	bl/rt	Mirror common	450	
W19	X2:17	X6:4	1	rt/ge	Mirror heater +	450	B
W20	SP1	X6:5	1	br/ge	Heater ground	480	B
W21	X2:18	X7:1	0.75	ge/rt	Speaker +, T2	420	
W22	X2:19	X7:2	0.75	ge/br	Speaker −, T2	420	

Ref	Function	Ways	Used	Area
X1	Body inline	16	4	Dry
X2	Door control unit	26	19	Dry
X3	Window motor	4	4	Sealed
X4	Door lock	6	4	Sealed
X5	Handle sensor	2	2	Sealed
X6	Mirror	6	5	Sealed
X7	Speaker	2	2	Sealed

Seg.	L mm	Wires	Ø mm	Protection
S1	140	4	5.5	Conduit NW7
S2	60	4	5.5	Tape full
S3	120	6	6.4	Tape full
S4	90	19	10.9	Tape spot
S5	260	12	8.4	Tape full
S6	150	4	5.5	Tape spot
S7	220	6	5.4	Tape full
S8	120	4	4.6	Conduit NW7
S9	80	2	2.7	Tape spot
S10	360	5	5.3	Conduit NW7
S11	70	2	3.2	Tape spot

VOLTAGE DROP CHECK, WINDOW MOTOR

Loop W01, W07, W08, W03, W02 at 20 °C: 13.8 mΩ. At 15 A * the drop is **0.21 V**, limit 0.60 V *. **Pass.**

METHOD

Bundle Ø = 1.2 × √(Σ wire OD²) *. Conduit NW is the smallest size at least 1.25 × Ø. Wire lengths are the sum of bundle segments on the route.

REVISIONS

REV	DESCRIPTION	DATE
A	First issue	2026-11-08
B	CR-0042: mirror heater added. New W19, W20; X6 cavities 4, 5; S3, S4 and S10 re-sized	2026-11-08

NOTES

1. Wire FLRY-B to ISO 6722-1, thin wall. Colours in German abbreviations: rt red, br brown, ge yellow, gn green, bl blue, vi violet, gr grey, or orange, sw black, ws white.
2. T1, T2: twisted pairs, lay length 25 ±5 mm.
3. Splice SP1: ultrasonic weld, sealed with adhesive-lined heat-shrink.
4. Bundle length tolerance ±5 mm up to 500 mm, ±10 mm above.
5. Tape full: 50 % overlap, PET cloth. Tape spot: 100 mm pitch. Conduit: slit PA corrugated, wet area of door.
6. Connectors in the door wet area are sealed. Grommet G1 sealed to the door skin.
7. * Assumed design values. Not built or tested.

Material: see notes	Standards: ISO 6722-1	Drawing type: formboard	Mass: -
Scale: NTS	Tolerancing principle: ISO 8015	Designed by Mahdi Bachraoui, 2026-11-08	Checked by: -
Document type: Harness drawing	Door harness Generic example, fictional vehicle		Format: A3
Drawing Number: MB-008			Rev B Sheet 1/1